

Lab Experiment 5: Locate and Interpret System Log Files

Objective

- To understand what **system log files** are and why they are important.
- To learn how to **locate**, **open**, and **read** log files in Linux.
- To identify different types of logs: system, kernel, authentication, and boot logs.

Background

- **Logs** are records of system and user activities.
- Linux stores logs inside the directory:

`/var/log/`

- Logs help in:
 - Detecting errors and warnings.
 - Monitoring login/security events.
 - Checking system boot and shutdown details.
 - Troubleshooting system performance issues.

Common log files in Linux:

Log File	Purpose
/var/log/syslog	General system messages
/var/log/dmesg	Kernel boot messages
/var/log/auth.log	Authentication and login activity
/var/log/kern.log	Kernel-specific messages
/var/log/boot.log	Boot and startup information
/var/log/apt/history.log	Package installation history
/var/log/Xorg.0.log	Graphics/display server log

Steps to Perform

Step 1: Navigate to Log Directory

1. Go to log folder:

```
cd /var/log
```

2. List log files:

```
ls
```

Step 2: View Log Files

3. View system log:

```
cat /var/log/syslog
```

4. View authentication log:

```
cat /var/log/auth.log
```

5. View kernel log:

```
cat /var/log/kern.log
```

6. View last 20 lines:

```
tail -20 /var/log/syslog
```

7. View logs in real-time:

```
tail -f /var/log/syslog
```

(Press Ctrl + C to stop)

Step 3: Read Boot and Filter Errors

8. View boot log:

```
cat /var/log/boot.log
```

9. Search errors from log:

```
grep "error" /var/log/syslog
```

10. Search warnings:

```
grep "warn" /var/log/syslog
```

Step 4: Check User Login Logs

11. View login activity:

```
cat /var/log/auth.log
```

12. See recent user logins:

```
last
```

13. Check failed login attempts:

```
grep "Failed" /var/log/auth.log
```

Step 5: Package Management Logs

14. View software installation history:

```
sudo cat /var/log/apt/history.log
```

15. Check package upgrade log:

```
sudo cat /var/log/apt/term.log
```

Step 6: Kernel and Boot Messages

16. View kernel messages:

```
sudo dmesg | less
```

If it shows empty use:

```
sudo journalctl -k
```

17. Filter USB events:

```
sudo dmesg | grep usb
```

18. Check boot messages:

```
sudo journalctl -b
```

Step 7: Log File Management

19. Check size of log file:

```
ls -lh /var/log/syslog
```

20. List compressed archived logs:

```
ls /var/log/*.gz
```

21. View compressed log:

```
zcat /var/log/syslog.1.gz
```

Outcome

- Students can locate Linux log files.
- Students can read and interpret system and security logs.
- Students can analyze logs for troubleshooting and security checks.

Viva Questions

1. What is a log file?
2. Where are Linux logs stored?
3. Which log records user authentication events?
4. Which command displays kernel messages?
5. How do you view live logs?
6. Why are logs important for system administration?
7. How to check recently installed software?
8. Which command displays last login information?
9. What is the function of **grep** in logs?
10. How can compressed logs be viewed?